

Statistics & Sampling

ANSWER KEY



Sampling methods

- 1 A school wants to survey students about the cafeteria. Explain why surveying only students in the chess club would produce biased results, and suggest a better sampling method.

Surveying only chess club members would not represent the full student body — their opinions might differ systematically from other students (e.g. different schedules, different cafeteria usage patterns), introducing bias. A better method: use random sampling, such as randomly selecting students from the full enrollment list across all grades and interests, so every student has an equal chance of being surveyed.

Comparing distributions using shape, center, and spread

- 2 Two data sets have the same median but Data Set A has a much larger range than Data Set B. Explain what this tells you about the variability of each data set.

Since Data Set A has a larger range despite having the same median, its data values are more spread out (more variable) than Data Set B's — the values in Data Set A extend further from the center, even though both sets are centered around the same typical value.

Linear regression and correlation

- 3 A scatterplot of hours studied vs. exam score has a line of best fit $y = 6.5x + 52$. Interpret the slope and y -intercept in context.

The slope, 6.5, means each additional hour of studying is associated with an increase of about 6.5 points on the exam score. The y -intercept, 52, represents the predicted exam score for a student who studies 0 hours.

- 4 A data set has a correlation coefficient of $r = -0.85$. Describe the strength and direction of the relationship.

Since r is negative, the relationship is a negative (inverse) association — as one variable increases, the other tends to decrease. Since $|r| = 0.85$ is close to 1, the relationship is strong.

Using a line of best fit to make predictions

- 5 A line of best fit for predicting weight (in kg) from height (in cm) is $W = 0.9H - 70$. Predict the weight of a person who is 175 cm tall, and explain a limitation of using this model for a person who is 250 cm tall.

$W = 0.9(175) - 70 = 157.5 - 70 = 87.5$ kg. Using the model for a height of 250 cm would be an extrapolation far outside the range of the original data used to build the model, so the prediction may be unreliable — the linear relationship observed for typical human heights may not hold at such an extreme, unrealistic value.

Interpreting the standard deviation

- 6 Data Set A has standard deviation 2, and Data Set B has standard deviation 8, both with the same mean. Explain what this indicates about the two data sets, and give a real-world example of each.

A smaller standard deviation (Data Set A) means the data values are clustered closely around the mean, while a larger standard deviation (Data Set B) means the values are more spread out. Example: test scores on an easy, uniform quiz might have a small standard deviation (most students score similarly), while test scores on a challenging exam with mixed preparation levels might have a large standard deviation (scores vary widely).

Margin of error and confidence in survey results

- 7 A survey estimates that 42% of a population favors a new policy, with a margin of error of 3%. A second survey with a larger sample size estimates the same statistic with a margin of error of 1.5%. Explain why the second survey's margin of error is smaller, and what this means for the precision of the estimate.

A larger sample size generally produces a smaller margin of error, because a bigger sample gives a more precise (less variable) estimate of the true population value. The second survey's narrower margin of error (1.5% vs. 3%) means its estimate is more precise — the true population percentage is likely confined to a tighter range around the reported statistic.

Evaluating experimental design

- 8 A researcher wants to test whether a new fertilizer increases crop yield. Explain why using a control group (plants without the fertilizer) is essential to this experiment's design.

A control group provides a baseline for comparison — without it, the researcher cannot determine whether any observed increase in crop yield is actually caused by the fertilizer, or whether it would have happened anyway due to other factors (weather, soil quality, time). Comparing the treatment group (fertilizer) to the control group (no fertilizer) under otherwise identical conditions isolates the fertilizer's specific effect.

Calculating and comparing means from grouped data

- 9 A class of 25 students has a mean test score of 78. A second class of 15 students has a mean score of 88. Find the combined mean score of all 40 students.

Total points from class 1: $25 \times 78 = 1950$. Total points from class 2: $15 \times 88 = 1320$. Combined total: $1950 + 1320 = 3270$. Combined mean: $\frac{3270}{40} = 81.75$.

Recognizing bias in survey questions and methods

- 10 A survey asks, "Don't you agree that the excellent new park should be funded?" Explain why this question is biased, and rewrite it in a neutral form.

The question is biased because it uses leading language ("Don't you agree" and "excellent") that pressures respondents toward a particular answer, rather than allowing them to form an independent opinion. A neutral version: "Do you support funding the new park? Yes or No."

Comparing two data sets using boxplot-style summaries

- 11 Data Set A has minimum 10, Q1 = 20, median 30, Q3 = 40, maximum 50. Data Set B has minimum 15, Q1 = 28, median 30, Q3 = 32, maximum 45. Both sets have the same median. Compare their spread using the interquartile range (IQR).

IQR of Set A: $Q3 - Q1 = 40 - 20 = 20$. IQR of Set B: $Q3 - Q1 = 32 - 28 = 4$. Since Set A has a much larger IQR, its middle 50% of data is far more spread out than Set B's, even though both sets share the same median.

Extrapolation vs. interpolation

- 12 A line of best fit is built from data on shoe size (ranging from 6 to 12) and height. Explain the difference between using this model to predict height for a shoe size of 9 versus a shoe size of 20, referencing interpolation and extrapolation.

Predicting height for a shoe size of 9 is interpolation, since 9 falls within the original data range (6 to 12) — this is generally reliable. Predicting for a shoe size of 20 is extrapolation, since it falls far outside the observed data range — this is much less reliable, since the linear relationship observed within the data range may not hold true for such an extreme, unobserved value.

Relative frequency and probability from a frequency table

- 13 A frequency table of exam grades shows: A: 8 students, B: 15 students, C: 12 students, D: 5 students. Find the relative frequency (as a percent) of students who earned a B or higher.

Total students: $8 + 15 + 12 + 5 = 40$. Students with B or higher: $8 + 15 = 23$. Relative frequency: $\frac{23}{40} = 0.575$, or 57.5%.

Random assignment vs. random sampling

- 14 Explain the difference between random sampling and random assignment, and state which one allows a study to make causal claims.

Random sampling refers to how participants are SELECTED from a population, which allows results to be generalized to that population. Random assignment refers to how selected participants are ASSIGNED to treatment vs. control groups, which allows a study to make causal claims — because random assignment balances out other confounding factors between groups, any resulting difference can be attributed to the treatment itself rather than pre-existing group differences.

A multi-part regression and residual analysis problem

- 15 This question has three parts. A line of best fit for predicting sales revenue (in thousands of dollars) from advertising spend (in thousands of dollars) is $R = 4.2A + 18$. (a) Predict the revenue for an advertising spend of 10 thousand dollars. (b) If the actual observed revenue at $A = 10$ was 55 thousand dollars, find the residual (actual minus predicted). (c) Explain what a positive residual indicates about this particular data point relative to the line of best fit.

(a) $R = 4.2(10) + 18 = 42 + 18 = 60$ thousand dollars predicted. (b) Residual = actual – predicted = $55 - 60 = -5$ thousand dollars. (c) Since the residual here is actually negative (-5), this indicates the actual revenue was lower than the model predicted — this specific data point lies below the line of best fit. A positive residual, in general, would indicate the actual value exceeds the model's prediction, meaning the point lies above the line.

Multiple choice — easy

- 16 A homeroom class of 27 students has a mean test score of 86. A second section of 38 students has a mean score of 96. Find the combined mean score of all 65 students, to the nearest hundredth.
- a. 91.00
 - b. 89.85
 - c. 93.85
 - d. 91.85 ✓

Total points from section 1: $27 \times 86 = 2322$. Total points from section 2: $38 \times 96 = 3648$. Combined total: $2322 + 3648 = 5970$. Combined mean: $5970/65 = 91.85$.

- 17 A chemistry class of 36 students has a mean test score of 92. A second section of 26 students has a mean score of 62. Find the combined mean score of all 62 students, to the nearest hundredth.
- a. 77.00
 - b. 79.42 ✓
 - c. 81.42
 - d. 77.42

Total points from section 1: $36 \times 92 = 3312$. Total points from section 2: $26 \times 62 = 1612$. Combined total: $3312 + 1612 = 4924$. Combined mean: $4924/62 = 79.42$.

- 18 A homeroom class of 40 students has a mean test score of 93. A second section of 26 students has a mean score of 91. Find the combined mean score of all 66 students, to the nearest hundredth.
- a. 90.21
 - b. 92.00
 - c. 92.21 ✓
 - d. 94.21

Total points from section 1: $40 \times 93 = 3720$. Total points from section 2: $26 \times 91 = 2366$. Combined total: $3720 + 2366 = 6086$. Combined mean: $6086/66 = 92.21$.

- 19 A history class of 15 students has a mean test score of 67. A second section of 21 students has a mean score of 73. Find the combined mean score of all 36 students, to the nearest hundredth.
- a. 70.00
 - b. 68.50
 - c. 72.50
 - d. 70.50 ✓

Total points from section 1: $15 \times 67 = 1005$. Total points from section 2: $21 \times 73 = 1533$. Combined total: $1005 + 1533 = 2538$. Combined mean: $2538/36 = 70.50$.

- 20 A history class of 20 students has a mean test score of 86. A second section of 29 students has a mean score of 96. Find the combined mean score of all 49 students, to the nearest hundredth.
- a. 89.92
 - b. 93.92
 - c. 91.00
 - d. 91.92 ✓

Total points from section 1: $20 \times 86 = 1720$. Total points from section 2: $29 \times 96 = 2784$. Combined total: $1720 + 2784 = 4504$. Combined mean: $4504/49 = 91.92$.

- 21 An algebra class of 32 students has a mean test score of 98. A second section of 13 students has a mean score of 71. Find the combined mean score of all 45 students, to the nearest hundredth.
- a. 92.20
 - b. 84.50
 - c. 90.20 ✓
 - d. 88.20

Total points from section 1: $32 \times 98 = 3136$. Total points from section 2: $13 \times 71 = 923$. Combined total: $3136 + 923 = 4059$. Combined mean: $4059/45 = 90.20$.

- 22 A biology class of 27 students has a mean test score of 95. A second section of 13 students has a mean score of 88. Find the combined mean score of all 40 students, to the nearest hundredth.

a. 92.72 ✓

b. 94.72

c. 91.50

d. 90.72

Total points from section 1: $27 \times 95 = 2565$. Total points from section 2: $13 \times 88 = 1144$. Combined total: $2565 + 1144 = 3709$. Combined mean: $3709/40 = 92.72$.

- 23 A chemistry class of 27 students has a mean test score of 75. A second section of 10 students has a mean score of 60. Find the combined mean score of all 37 students, to the nearest hundredth.

a. 67.50

b. 72.95

c. 68.95

d. 70.95 ✓

Total points from section 1: $27 \times 75 = 2025$. Total points from section 2: $10 \times 60 = 600$. Combined total: $2025 + 600 = 2625$. Combined mean: $2625/37 = 70.95$.

- 24 An algebra class of 30 students has a mean test score of 81. A second section of 18 students has a mean score of 98. Find the combined mean score of all 48 students, to the nearest hundredth.

a. 89.38

b. 89.50

c. 85.38

d. 87.38 ✓

Total points from section 1: $30 \times 81 = 2430$. Total points from section 2: $18 \times 98 = 1764$. Combined total: $2430 + 1764 = 4194$. Combined mean: $4194/48 = 87.38$.

- 25 A history class of 29 students has a mean test score of 92. A second section of 10 students has a mean score of 67. Find the combined mean score of all 39 students, to the nearest hundredth.

a. 85.59 ✓

b. 83.59

c. 87.59

d. 79.50

Total points from section 1: $29 \times 92 = 2668$. Total points from section 2: $10 \times 67 = 670$. Combined total: $2668 + 670 = 3338$. Combined mean: $3338/39 = 85.59$.

- 26 An algebra class of 32 students has a mean test score of 83. A second section of 25 students has a mean score of 91. Find the combined mean score of all 57 students, to the nearest hundredth.

a. 88.51

b. 84.51

c. 87.00

d. 86.51 ✓

Total points from section 1: $32 \times 83 = 2656$. Total points from section 2: $25 \times 91 = 2275$. Combined total: $2656 + 2275 = 4931$. Combined mean: $4931/57 = 86.51$.

- 27 A homeroom class of 12 students has a mean test score of 62. A second section of 14 students has a mean score of 78. Find the combined mean score of all 26 students, to the nearest hundredth.

a. 70.00

b. 72.62

c. 68.62

d. 70.62 ✓

Total points from section 1: $12 \times 62 = 744$. Total points from section 2: $14 \times 78 = 1092$. Combined total: $744 + 1092 = 1836$. Combined mean: $1836/26 = 70.62$.

- 28 A homeroom class of 35 students has a mean test score of 76. A second section of 26 students has a mean score of 60. Find the combined mean score of all 61 students, to the nearest hundredth.

a. 68.00
b. 67.18
c. 71.18
d. 69.18 ✓

Total points from section 1: $35 \times 76 = 2660$. Total points from section 2: $26 \times 60 = 1560$. Combined total: $2660 + 1560 = 4220$. Combined mean: $4220/61 = 69.18$.

- 29 A homeroom class of 17 students has a mean test score of 63. A second section of 29 students has a mean score of 74. Find the combined mean score of all 46 students, to the nearest hundredth.

a. 68.50
b. 67.93
c. 71.93
d. 69.93 ✓

Total points from section 1: $17 \times 63 = 1071$. Total points from section 2: $29 \times 74 = 2146$. Combined total: $1071 + 2146 = 3217$. Combined mean: $3217/46 = 69.93$.

- 30 An algebra class of 10 students has a mean test score of 82. A second section of 10 students has a mean score of 73. Find the combined mean score of all 20 students, to the nearest hundredth.

a. 78.50
b. 75.50
c. 77.50 ✓
d. 79.50

Total points from section 1: $10 \times 82 = 820$. Total points from section 2: $10 \times 73 = 730$. Combined total: $820 + 730 = 1550$. Combined mean: $1550/20 = 77.50$.

Multiple choice — medium

- 31** A line of best fit for predicting race time (minutes) from training hours is $y = -0.50x - 15$. Predict the value of y when $x = 5$.

a. -17.50 ✓

b. -18.00

c. -19.50

d. -15.50

$$y = -0.50(5) - 15 = -17.50.$$

- 32** A line of best fit for predicting sales revenue (thousands of dollars) from advertising spend (thousands of dollars) is $y = 4.00x + 62$. Predict the value of y when $x = 18$.

a. 134.00 ✓

b. 132.00

c. 138.00

d. 136.00

$$y = 4.00(18) + 62 = 134.00.$$

- 33** A line of best fit for predicting exam score from hours studied is $y = -0.50x + 41$. Predict the value of y when $x = 2$.

a. 42.00

b. 39.50

c. 40.00 ✓

d. 38.00

$$y = -0.50(2) + 41 = 40.00.$$

- 34** A line of best fit for predicting ice cream sales (dollars) from temperature (degrees F) is $y = -4.50x + 50$. Predict the value of y when $x = 18$.

- a. -35.50
- b. -29.00
- c. -33.00
- d. -31.00 ✓

$$y = -4.50(18) + 50 = -31.00.$$

- 35** A line of best fit for predicting salary (thousands of dollars) from years of experience is $y = -1.50x + 59$. Predict the value of y when $x = 20$.

- a. 27.00
- b. 31.00
- c. 27.50
- d. 29.00 ✓

$$y = -1.50(20) + 59 = 29.00.$$

- 36** A line of best fit for predicting sales revenue (thousands of dollars) from advertising spend (thousands of dollars) is $y = 4.50x + 7$. Predict the value of y when $x = 18$.

- a. 92.50
- b. 90.00
- c. 86.00
- d. 88.00 ✓

$$y = 4.50(18) + 7 = 88.00.$$

- 37** A line of best fit for predicting salary (thousands of dollars) from years of experience is $y = -1.00x - 26$. Predict the value of y when $x = 2$.

a. -28.00 ✓

b. -29.00

c. -30.00

d. -26.00

$$y = -1.00(2) - 26 = -28.00.$$

- 38** A line of best fit for predicting salary (thousands of dollars) from years of experience is $y = -4.00x + 38$. Predict the value of y when $x = 11$.

a. -4.00

b. -8.00

c. -10.00

d. -6.00 ✓

$$y = -4.00(11) + 38 = -6.00.$$

- 39** A line of best fit for predicting race time (minutes) from training hours is $y = -1.50x + 10$. Predict the value of y when $x = 18$.

a. -17.00 ✓

b. -15.00

c. -18.50

d. -19.00

$$y = -1.50(18) + 10 = -17.00.$$

- 40 A line of best fit for predicting race time (minutes) from training hours is $y = 1.50x - 12$. Predict the value of y when $x = 12$.
- a. 4.00
 - b. 8.00
 - c. 6.00 ✓
 - d. 7.50

$$y = 1.50(12) - 12 = 6.00.$$

- 41 A line of best fit for predicting exam score from hours studied is $y = -2.00x + 72$. Predict the value of y when $x = 11$.
- a. 51.00
 - b. 48.00
 - c. 52.00
 - d. 50.00 ✓

$$y = -2.00(11) + 72 = 50.00.$$

- 42 A line of best fit for predicting sales revenue (thousands of dollars) from advertising spend (thousands of dollars) is $y = -2.50x + 31$. Predict the value of y when $x = 12$.
- a. -1.00
 - b. -1.50
 - c. 3.00
 - d. 1.00 ✓

$$y = -2.50(12) + 31 = 1.00.$$

- 43 A line of best fit for predicting ice cream sales (dollars) from temperature (degrees F) is $y = -3.50x + 22$. Predict the value of y when $x = 5$.

a. 4.50 ✓

b. 1.00

c. 2.50

d. 6.50

$$y = -3.50(5) + 22 = 4.50.$$

- 44 A line of best fit for predicting exam score from hours studied is $y = -4.00x + 24$. Predict the value of y when $x = 6$.

a. -2.00

b. 2.00

c. 0.00 ✓

d. -4.00

$$y = -4.00(6) + 24 = 0.00.$$

- 45 A line of best fit for predicting race time (minutes) from training hours is $y = -3.00x + 35$. Predict the value of y when $x = 4$.

a. 23.00 ✓

b. 20.00

c. 21.00

d. 25.00

$$y = -3.00(4) + 35 = 23.00.$$

Multiple choice — hard

- 46 A data set has minimum 9, $Q1 = 17$, median 25, $Q3 = 31$, maximum 44. Find the interquartile range (IQR).

a. 14 ✓

b. 16

c. 35

d. 12

$$IQR = Q3 - Q1 = 31 - 17 = 14.$$

- 47 A data set has minimum 19, $Q1 = 23$, median 47, $Q3 = 55$, maximum 65. Find the interquartile range (IQR).

a. 46

b. 34

c. 30

d. 32 ✓

$$IQR = Q3 - Q1 = 55 - 23 = 32.$$

- 48 A data set has minimum 24, $Q1 = 35$, median 52, $Q3 = 71$, maximum 74. Find the interquartile range (IQR).

a. 34

b. 38

c. 50

d. 36 ✓

$$IQR = Q3 - Q1 = 71 - 35 = 36.$$

- 49 A data set has minimum 28, $Q1 = 38$, median 54, $Q3 = 75$, maximum 84. Find the interquartile range (IQR).

a. 37 ✓

b. 35

c. 56

d. 39

$$IQR = Q3 - Q1 = 75 - 38 = 37.$$

- 50 A data set has minimum 6, $Q1 = 12$, median 17, $Q3 = 18$, maximum 23. Find the interquartile range (IQR).

a. 4

b. 6 ✓

c. 17

d. 8

$$IQR = Q3 - Q1 = 18 - 12 = 6.$$

- 51 A line of best fit is $y = 4x + 36$. At $x = 9$, the actual observed value is $y = 64$. Find the residual (actual minus predicted).

a. -10

b. -6

c. 72

d. -8 ✓

$$\text{Predicted value: } y = 4(9) + 36 = 72. \text{ Residual} = 64 - (72) = -8.$$

- 52 A line of best fit is $y = -8x + 5$. At $x = 6$, the actual observed value is $y = -47$. Find the residual (actual minus predicted).
- a. -4 ✓
 - b. -6
 - c. -2
 - d. -43

Predicted value: $y = -8(6) + 5 = -43$. Residual $= -47 - (-43) = -4$.

- 53 A line of best fit is $y = 6x + 7$. At $x = 17$, the actual observed value is $y = 114$. Find the residual (actual minus predicted).
- a. 109
 - b. 5 ✓
 - c. 3
 - d. 7

Predicted value: $y = 6(17) + 7 = 109$. Residual $= 114 - (109) = 5$.

- 54 A line of best fit is $y = -5x + 8$. At $x = 12$, the actual observed value is $y = -49$. Find the residual (actual minus predicted).
- a. 3 ✓
 - b. 1
 - c. 5
 - d. -52

Predicted value: $y = -5(12) + 8 = -52$. Residual $= -49 - (-52) = 3$.

- 55 A line of best fit is $y = -2x + 13$. At $x = 11$, the actual observed value is $y = -13$. Find the residual (actual minus predicted).

- a. -6
- b. -4 ✓
- c. -9
- d. -2

Predicted value: $y = -2(11) + 13 = -9$. Residual $= -13 - (-9) = -4$.

- 56 A data set has $Q1 = 11$ and $Q3 = 21$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.

- a. 21
- b. 38
- c. 36 ✓
- d. 34

$IQR = Q3 - Q1 = 21 - 11 = 10$. Upper fence $= Q3 + 1.5 \times IQR = 21 + 1.5(10) = 36$.

- 57 A data set has $Q1 = 21$ and $Q3 = 49$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.

- a. 89
- b. 49
- c. 91 ✓
- d. 93

$IQR = Q3 - Q1 = 49 - 21 = 28$. Upper fence $= Q3 + 1.5 \times IQR = 49 + 1.5(28) = 91$.

58 A data set has $Q1 = 22$ and $Q3 = 60$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.

- a. 60
- b. 119
- c. 117 ✓
- d. 115

$$IQR = Q3 - Q1 = 60 - 22 = 38. \text{ Upper fence} = Q3 + 1.5 \times IQR = 60 + 1.5(38) = 117.$$

59 A data set has $Q1 = 37$ and $Q3 = 43$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.

- a. 54
- b. 50
- c. 43
- d. 52 ✓

$$IQR = Q3 - Q1 = 43 - 37 = 6. \text{ Upper fence} = Q3 + 1.5 \times IQR = 43 + 1.5(6) = 52.$$

60 A data set has $Q1 = 40$ and $Q3 = 50$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.

- a. 67
- b. 65 ✓
- c. 50
- d. 63

$$IQR = Q3 - Q1 = 50 - 40 = 10. \text{ Upper fence} = Q3 + 1.5 \times IQR = 50 + 1.5(10) = 65.$$